

Dreamory: AI-Powered Bedtime Storytelling for Emotional Reframing Before In-Sleep Memory Consolidation

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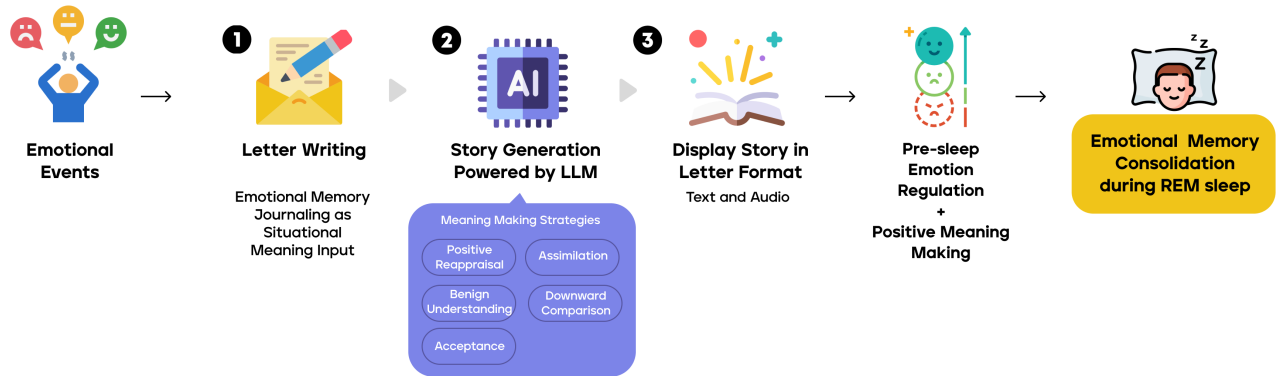


Figure 1: Dreamory's User Flow for Emotional Memory Consolidation through the Meaning-Making Model

Abstract

This paper presents Dreamory, an AI-powered storytelling system designed to influence emotional memory consolidation during sleep through pre-sleep narrative interventions. Drawing from the Meaning-Making Model, Dreamory transforms users' daily emotional experiences into personalized bedtime stories that promote positive reframing. The system enables users to journal their emotional experiences via letter writing and leverages a large language model (LLM) to analyze emotional content and generate narratives with meaning-making strategies. A one-week pilot study with four participants demonstrated Dreamory's potential to enhance pre-sleep emotional regulation and foster positive reframing of experiences. Participants also reported a perceived improvement in self-efficacy and sleep quality. This work proposes an inspiring approach that leverages generative AI for positive meaning-making of emotional memory, promoting pre-sleep emotional shifts, fostering

positive mindsets, and enhancing emotional memory consolidation to support mental well-being.

CCS Concepts

• **Human-centered computing** → **Interaction design**; *Empirical studies in HCI*.

Keywords

AI-Powered Interventions, Personalized Storytelling, Emotional Memory Consolidation, Pre-Sleep Emotional Regulation, Meaning-Making, Narrative Therapy

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1 Introduction

Emotional memory consolidation during sleep plays a crucial role in mental health and well-being [12, 22]. Research has shown that negative emotional experiences are preferentially consolidated during REM sleep, reinforcing distressing emotions and embedding them into long-term memory [19]. This process can perpetuate a cycle of stress and anxiety, highlighting the need for interventions that regulate emotional memory before sleep [22]. While traditional interventions like mindfulness and cognitive behavioral therapy

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effectively regulate emotions [13], they rarely target the pre-sleep period. Research highlights this pre-sleep window as a unique opportunity to shape how emotional experiences are processed and integrated into autobiographical memory [8, 23]. Addressing the emotional memories before sleep may significantly impact their consolidation, potentially disrupting the reinforcement of negative emotions [8].

In this work, we drew from the Meaning-Making Model [14] to develop a targeted pre-sleep intervention for emotional memory reconsolidation through storytelling [7, 16]. We present Dreamory, an AI-powered system that transforms users' daily emotional experiences, captured through letter-writing, into personalized bedtime narratives that promote positive reframing. Through this process, Dreamory helps users reinterpret their emotional experiences before sleep, potentially influencing how these memories are consolidated during subsequent sleep phases.

Our pilot study with four university students demonstrated Dreamory's potential to enhance pre-sleep emotional regulation and memory reconsolidation. Quantitative results showed improvements in emotional state and the intensity and valence of their emotional memories, indicating more positive reinterpretations. Participants also reported increased self-efficacy, optimism, and sleep quality. Qualitative feedback highlighted that Dreamory facilitated smoother emotional processing before sleep, with participants noting that the system helped them relax, fall asleep more easily, and effectively reflect on and reframe their daily experiences.

This work makes three primary contributions. First, we present Dreamory, an AI-powered bedtime storytelling system that leverages meaning-making principles to guide emotional memory reconsolidation before sleep. Second, through a pilot study, we provide empirical evidence demonstrating how AI-guided narrative interventions can positively impact emotional well-being, including improvements in emotional regulation, memory processing, and sleep quality. Finally, through exploring how AI-generated narratives can influence emotional memory consolidation, we highlight a novel design space of pre-sleep memory augmentation. This work provides HCI researchers with new perspectives on memory interventions, inspires sleep technology designers to innovate in enhancing sleep-related well-being, and highlights opportunities for affective interaction designers to utilize the pre-sleep period for emotional support.

2 Related work

2.1 Sleep Technology in HCI and Emotion Consolidation During Sleep

Early work in Human-Computer Interaction (HCI) explored opportunities for unobtrusive sleep technologies that integrate tracking, relaxation, and educational tools to support healthy sleep behaviors [4]. Building on this foundation, researchers have developed interventions targeting both in-sleep and pre-sleep stages. For example, Semertzidis et al. introduced neurofeedback systems like Inter-Dream [21] and Dozer [20] to reduce cognitive arousal and promote pre-sleep relaxation. Similarly, Amores et al. [1] developed Ezzence, a modular scent wearable that uses EEG data to deliver targeted scents during specific sleep stages, improving sleep quality and memory consolidation. Li et al. [10] presented SleepyFlora, an

interactive system that enables a user to know the other user's sleep states and augment their sleep by sending soothing aromas and playing music remotely. Zhang et al. [24] demonstrated the effectiveness of combining spatio-temporal tactile displays with music to reduce pre-sleep stress, while Kapoor [9] highlighted the potential of tactile feedback for resonant breathing to enhance relaxation and sleep onset. Recent work has also explored post-sleep interactions. Lyu et al. [11] presented LumaDreams which supports positive dream transformation using generative AI and demonstrates improvements in sleep quality and daily empowerment.

While these technological interventions have shown promise in promoting relaxation and sleep onset, they often overlook sleep's crucial role in emotional memory processing. Research has demonstrated that sleep, particularly REM and slow-wave sleep (SWS), plays a vital role in consolidating emotional memories and integrating emotionally charged experiences into long-term memory. In recent years, the relationship between sleep and emotional well-being has gained attention. Technologies promoting better sleep hygiene or pre-sleep relaxation through sensory stimuli, such as music or olfactory cues, have been shown to provide short-term mood improvements [1, 24]. However, most of these efforts focus on immediate relaxation, overlooking deeper emotional outcomes. Addressing this gap, our work targets emotional memory optimization, exploring how sleep technologies can support long-term emotional well-being through in-sleep memory consolidation and emotion regulation.

Sleep plays a vital role in consolidating emotional memories by stabilizing and integrating emotionally charged experiences into long-term memory networks [18, 19]. REM sleep is crucial for processing negative emotional memories through theta-band interactions between the medial prefrontal cortex and limbic structures [5, 19]. Similarly, slow-wave sleep (SWS) supports memory consolidation through mechanisms such as Targeted Memory Reactivation (TMR), where sensory cues during sleep enhance memory processing [3].

Studies show that SWS preferentially consolidates negative memories over neutral ones, reflecting an evolutionary advantage for processing emotionally significant events [5, 17]. Additionally, TMR demonstrates the potential to influence emotional memory consolidation by re-exposing individuals to sensory cues during SWS, aiding in emotional regulation and resilience.

Leveraging these mechanisms presents unique opportunities for HCI systems to address emotional challenges. Pre-sleep journaling or narrative-based systems, like Dreamory, can help reframe emotional experiences, promoting regulation and reducing distress during REM and SWS. Such approaches have applications for conditions like PTSD and can enhance long-term emotional well-being [19]. By focusing on pre-sleep interventions and leveraging sleep's role in emotional memory processing, Dreamory aims to enhance emotional resilience and support positive memory consolidation.

2.2 Meaning Making and Storytelling

The Meaning-Making Model (MMM) [15] emphasizes the importance of reframing distressing events to align situational appraisals with global meaning systems, thereby fostering emotional regulation and psychological well-being. Common meaning-making

strategies include positive reappraisal—identifying beneficial aspects of challenges—and downward comparison, which emphasizes resilience by contrasting one’s experiences with more difficult scenarios. Empirical evidence demonstrates that positive reappraisal is associated with improved mood and reduced stress over time, while meaning-making efforts have been shown to help individuals integrate traumatic experiences into a coherent sense of self, reducing negative emotional impacts [6, 14].

Storytelling provides concrete techniques to facilitate the meaning-making process. Narrative therapy techniques, for instance, employs methods such as externalizing problems, deconstructing dominant narratives, and re-authoring life stories to reshape individual experiences. These approaches enable individuals to separate their identities from challenges, fostering resilience and agency [2]. Therapeutic letter writing, a key tool in narrative therapy, encourages reflection and emotional reconciliation, aligning closely with Dreamory’s letter-based design.

Dreamory integrates the principles of meaning-making and narrative therapy techniques into an AI-driven system. By analyzing emotional inputs through letter writing, the system generates empathetic, personalized narratives that support emotional reframing. Through its storytelling-based framework, Dreamory operationalizes meaning-making strategies, empowering users to reinterpret their experiences and foster emotional well-being.

3 System Design

Dreamory is an AI-driven storytelling system that combines meaning-making strategies with personalized narratives to influence emotional memory consolidation before sleep. The system operates through three key stages: letter writing, story generation, and story display.

Stage 1: Letter Writing. Dreamory begins by asking users to write a letter to journal their emotional memories. This was inspired by narrative therapy which starts from a past self expressing lived experiences, with therapeutic letter writing as a common way. In our work, to better support emotional memory journaling, we implemented a hint assistant that generates real-time prompts every five seconds based on user input. These prompts might ask users to elaborate on specific emotions (“How did you feel at that moment?”), (“What does this experience mean to you now?”) The letter format provides a natural structure for users to document their autobiographic memory and associated emotions.

Stage 2: Story Generation. After the user submits their letter, the system displays an envelope-sending animation with engaging messages such as “Are you ready for your unique story?” During this process, the LLM analyzes the letter to extract key events and emotional elements, then generates a personalized story using meaning-making strategies [15] including 1) Positive Reappraisal: Finding beneficial aspects in challenging situations; 2) Benign Understanding: Offering a compassionate or neutral perspective on distressing events; 3) Assimilation: Aligning situational experiences with the user’s broader personal meaning system to create coherence; 4) Downward comparison: Highlighting resilience by contrasting the user’s challenges with hypothetically more difficult scenarios; and 5) Acceptance: Encouraging acknowledgment and integration of past experiences as part of personal growth.

Stage 3: Story Display. The system presents the generated story in a letter format, beginning with “Dear [User’s Name],” and including a brief reflective comment preceding the narrative. To enrich the experience, the system also provides an audio version of the story. Users can choose to read or listen to their personalized narratives. Users can navigate forward or backward through the audio, providing an accessible way to engage with the personalized narratives.

3.1 Implementation

Dreamory is implemented using SwiftUI and integrates two GPT-4o-mini assistants to facilitate emotion understanding and personalized narrative generation. The system operates through three key stages: interactive letter writing, emotional analysis, and story generation.

Interactive letter writing: Users begin by journaling their emotional experiences, guided by prompts generated by the first GPT-4o-mini assistant. This assistant acts as a gentle and empathetic letter writing companion, using warm and supportive language to encourage deep reflection. The prompts are dynamically updated in real-time, aligning with narrative therapy’s externalization framework, where users articulate and separate their emotions from their sense of self, reducing emotional burdens. For example, questions like “How did that moment make you feel?” or “Would you like to share more about that experience?” create a safe and inviting environment for emotional expression. Detailed prompt design are included in the Appendix.

Emotional Analysis: Once the journaling is complete, the second GPT-4o-mini assistant analyzes the overall emotional tone of the input. This process involves identifying key emotional elements, such as the intensity and polarity (positive or negative) of emotions expressed in the letter. By extracting and interpreting these emotional cues, the system gains a deeper understanding of the user’s emotional state, allowing it to tailor the subsequent story generation to address the user’s unique experiences and emotional needs effectively. This step ensures that the generated narrative aligns with the user’s emotions and fosters meaningful emotional engagement.

Story Generation: The personalized story is generated using OpenAI’s GPT-4o-mini, leveraging its advanced capabilities to craft narratives that emphasize resilience, personal growth, and emotional reframing. The process utilizes carefully designed prompts to ensure the story aligns with the user’s emotional analysis. These prompts incorporate meaning-making strategies, such as positive reappraisal, benign understanding, and acceptance and growth, reframing challenges into opportunities for personal development and healing.

Key considerations in story generation include maintaining an uplifting tone, aligning the narrative with the user’s unique emotional context, and addressing all major challenges outlined in their letters. The output is tailored to promote emotional relaxation and prepare the user for sleep-dependent memory consolidation. To enhance engagement and accessibility, the story is delivered both as text and through an optional voice-over, created using OpenAI’s Alloy voice in a text-to-speech (TTS) module¹. The voice-over features a warm, soothing, and gender-neutral tone, further supporting

¹<https://platform.openai.com/docs/guides/speech>

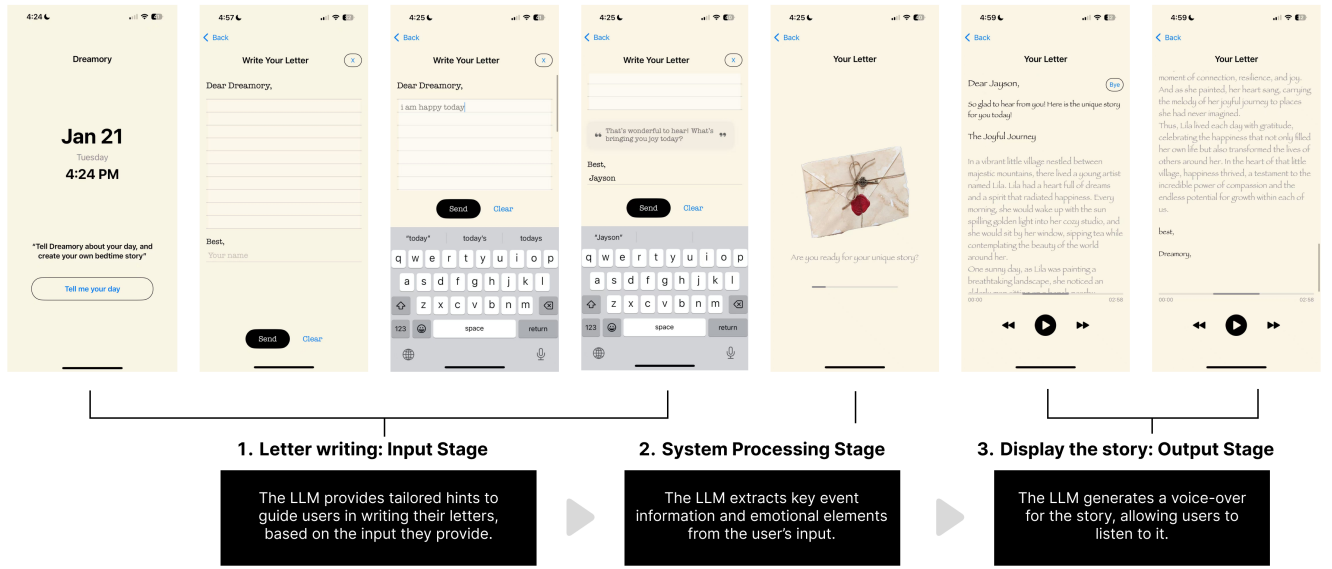


Figure 2: The Interaction Flow of Dreamory's User Interface.

emotional well-being and pre-sleep preparation. Detailed prompt designs are included in the Appendix.

4 Evaluation of Dreamory

With the approval of the local ethics committee, we conducted a week-long pilot study with four participants (all males, self-reported, Mean age = 21.75, SD = 1.299) to evaluate Dreamory's user experience in supporting emotional regulation and memory processing before sleep. To improve transparency, we labeled qualitative feedback with participant IDs (P1-P4). All participants demonstrated moderate to high familiarity with generative AI, with 100% reporting daily usage and half using such technologies multiple times per day. However, over half of the participants (P1,3,4) indicated they had never used sleep-related apps prior to this study.

During the experiment, participants accessed the Dreamory system through their personal smartphones by downloading the app in developer mode from a designated website. They engaged with the system one hour before their typical bedtime, within their usual sleep environments, following its intended use for pre-sleep reflection and relaxation. This setup allowed us to observe how users integrated Dreamory into their nightly routines and assessed its usability and effectiveness in facilitating emotional regulation and memory reframing.

We administered digital questionnaires before and after the experiment. All participants also participated in face-to-face interviews.

4.1 Experiment Design and Procedure

During the one-week intervention, participants engaged with the app daily and, upon completion, were asked to fill out a post-study survey. The pre- and post-surveys were designed to assess the system's impact on key factors, including emotional regulation

(Positive and Negative Affect Schedule - Short Form, PANAS-SF), memory processing (Memory Experiences Questionnaire, MEQ), self-efficacy (General Self-Efficacy Scale, GSE), optimism (Revised Life Orientation Test, LOT-R), and sleep quality (Pittsburgh Sleep Quality Index, PSQI). Following the study, we conducted 30-minute semi-structured interviews with each participant to understand their experience with Dreamory, including emotional processing through letter writing, the impact of AI-generated narratives on their pre-sleep emotions, and how the system influenced their emotional memory reconsolidation. Each participant received \$5 compensation.

4.1.1 Quantitative Results. Preliminary quantitative findings suggest that Dreamory has potential benefits in emotional regulation, memory engagement, and sleep quality. PANAS-SF scores showed a significant increase in positive affect (27.0 to 34.2) and a reduction in negative affect (18.9 to 15.5) (Figure 3). MEQ scores indicated stronger engagement with autobiographical memories (Emotional Intensity: 0.0 to 4.4) and a positive shift in memory interpretation (Valence: -4.0 to 1.3) (Figure 4). Self-efficacy and optimism improved as GSE scores rose from 29.25 to 30.50, and LOT-R scores increased from 9.5 to 11.25 (Figures 5, 6). Sleep quality showed improvement, with PSQI scores decreasing from 13.14 to 8.29, indicating reduced sleep latency and better overall sleep (Figure 7). In terms of usability, participants provided generally positive feedback, with high ratings for clarity and supportiveness ("Overwhelming vs Clear" at 6.25 and "Obstacle vs Assistance" at 5.25 on a 1–7 scale). However, moderate ratings for "Complex vs Simple" (4.0) signaled potential challenges with the app's usability.

4.1.2 Qualitative Results. Qualitative interview results revealed that Dreamory effectively supported emotional regulation, relaxation, and smoother transitions before sleep, seamlessly integrating

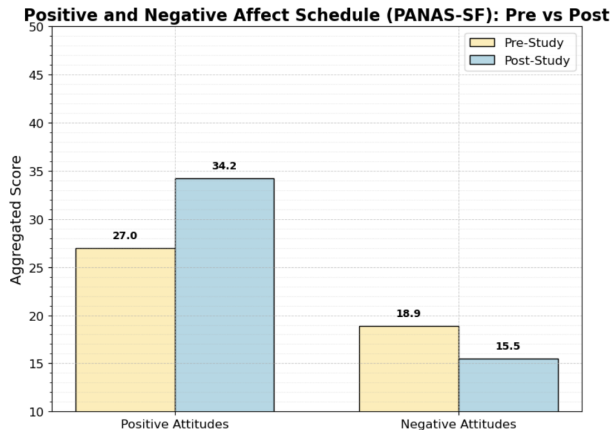


Figure 3: Aggregated PANAS-SF scores before and after the intervention.

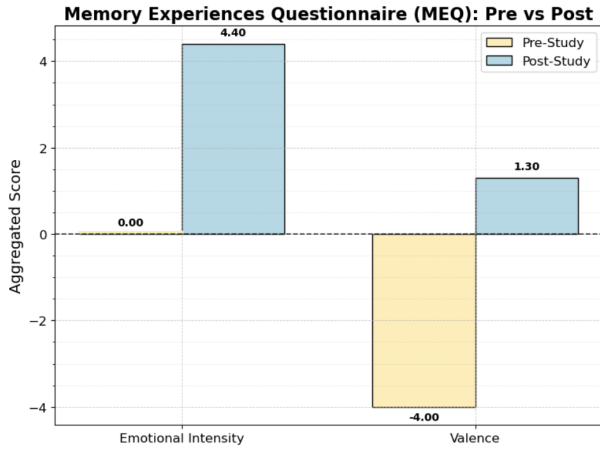


Figure 4: Aggregated MEQ scores for Emotional Intensity and Valence.

into participants' bedtime routines. Participants (P1, P2, P3, P4) praised the engaging and artistic tone of the stories and described the letter-writing process as a reflective and meaningful way to conclude the day.

One participant (P2) likened it to creating an “emotional parachute” that facilitated relaxation and prepared them for sleep. Additionally, Dreamory's storytelling for meaning-making was commended for transforming emotional inputs into constructive narratives. P2 and P3 also noted that the intervention inspired proactive action the following week, demonstrating its potential to align situational meaning with global meaning, fostering self-esteem and resilience.

All four participants found Dreamory intuitive and easy to use, with its interface fitting naturally into their nightly routines. However, participants suggested several improvements for future iterations:

- **Broader story settings (P2, P3):** Participants suggested incorporating imaginative scenarios, such as “Star Wars”

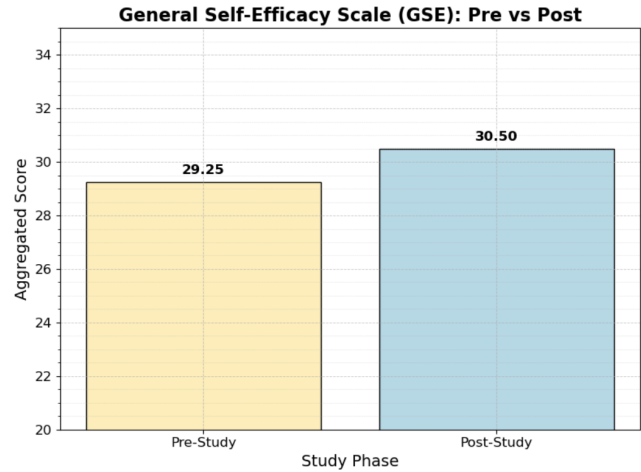


Figure 5: Aggregated GSE scores before and after the intervention.

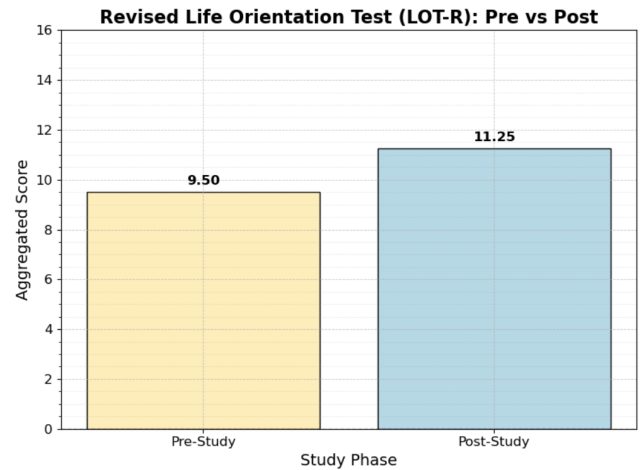


Figure 6: Aggregated LOT-R scores before and after the intervention.

or mythical tales, rather than solely reflecting real-world contexts.

- **Enhanced character traits (P2, P4):** They recommended making the letter recipient more human-like, with a clear personality that users could customize to foster a stronger emotional connection.
- **Long-term tracking (P2, P3):** They proposed adding a database to track interactions and recall users' previous input to reinforce continuity and deepen engagement over time.
- **Improved user experience (P1, P3, P4):** Suggestions included a loading progress bar (P3), a music playback bar (P1), simplifying the signature format (P1), and allowing users to access past letters and generated stories within the app (P4).
- **Expanded UI design improvements (P4):** P4 suggested introducing a day/night mode toggle, refining the color scheme,

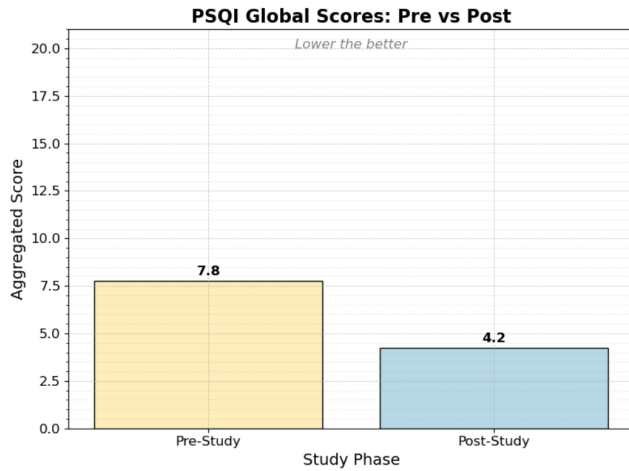


Figure 7: Aggregated PSQI scores before and after the intervention.

and making the interface more visually appealing to enhance engagement.

In summary, Dreamory demonstrated promise in enhancing emotional well-being, memory processing, and sleep quality, while identifying opportunities for usability, personalization, and feature enhancements in future iterations.

5 Limitations and Future Work

We acknowledge that our pilot study’s limited participant pool restricts comprehensive understanding of Dreamory’s user experience. Nevertheless, our initial exploration aimed to evaluate the feasibility, usability, and preliminary impact of Dreamory’s AI-driven bedtime storytelling approach, providing insights for design revisions and preparation for the formal user study.

This small-scale evaluation identified key improvement areas in both system design and functionality to enhance user engagement and effectiveness. From a design perspective, we plan to implement features such as a progress indicator to provide users with a clear understanding of their session flow, archiving past interactions for future reflection, and incorporating a distinct character to create a more immersive and personalized user experience. From a technical standpoint, addressing usability concerns such as slow loading times and enhancing the hint system during the journaling process will ensure smoother interactions. We also aim to integrate long-term tracking features with personalized feedback mechanisms, allowing users to monitor their progress over time. Future iterations of Dreamory will explore more structured retrieval-augmented generation (RAG) approaches or fine-tuned constraints to further enhance content fidelity to mitigate potential AI hallucinations.

After iterating the prototype of Dreamory, we will conduct a formal two-week study with 25 participants. Data analysis will include both quantitative and qualitative methods to provide a holistic understanding of Dreamory’s effectiveness. Pre- and post-intervention survey data will be analyzed using a paired-sample t-test to assess improvements in emotional regulation, self-efficacy,

optimism, and sleep quality. Qualitative data from semi-structured interviews will be thematically coded to identify user perceptions, emotional experiences, and engagement patterns. We aim to gain a deeper understanding of how users engage with Dreamory, evaluate its impact on emotional regulation and sleep quality, and derive actionable design implications for enhancing AI-driven bedtime storytelling systems.

6 Conclusion

This work introduced Dreamory, an AI-powered storytelling system designed to support emotional memory consolidation and well-being through pre-sleep narrative interventions. Through a one-week pilot study, we demonstrated its potential to enhance emotional regulation, self-efficacy, and sleep quality while identifying areas for improvement. By bridging meaning-making model through narrative therapy techniques with generative AI, this study offers a novel framework for designing systems that foster emotional resilience and inspire innovative approaches to integrating AI into mental health interventions.

Acknowledgments

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7 Appendices

A Interactive Journaling Prompt Design

The journaling phase employs carefully designed prompts to encourage users to articulate and reflect on their emotions deeply. These prompts, generated dynamically by the GPT-4o-mini assistant, align with narrative therapy principles to foster emotional externalization and self-awareness.

A.1 Guidelines for Journaling Prompts

- Use warm, supportive, and conversational tones.
- Avoid direct commands or imperatives to maintain a gentle approach.
- Focus on emotions and personal experiences to encourage deeper reflection.
- Dynamically adapt prompts in real-time based on user input for contextual relevance.

A.2 Examples of Journaling Prompts

- "How did that moment make you feel?"

- "Would you like to share more about those feelings?"
- "What thoughts crossed your mind during that experience?"
- "How do you think this experience has affected you?"

These prompts are designed to create a safe and empathetic space for users, enabling them to separate their challenges from their sense of self and reducing emotional burdens.

B Story Generation Prompt Design

The story generation phase utilizes structured prompts to craft personalized and uplifting narratives. The second GPT-4o-mini assistant integrates meaning-making strategies to reframe emotional experiences positively and align the story with the user's context.

B.1 Story Generation Prompts

You are a meaning-making story generator that uses the Meaning-Making Model (MMM) to transform emotional summaries into uplifting and healing stories. Your role is to take the user letter content and create a personalized story that emphasizes resilience, personal growth, and positive reinterpretation. Reframe negative experiences constructively, highlighting the user's journey toward overcoming challenges.

Output Structure:

- Title of the Story: [title]
- Story Content: [story]

- Maintain an uplifting, empathetic tone to promote emotional relaxation and well-being.
- Address all negative aspects in the emotional summary and reframe them positively, steering the narrative toward resilience and growth.
- Incorporate the following meaning-making strategies:
 - **Positive Reappraisal:** Highlight growth opportunities and beneficial aspects of challenging experiences. **Example Output in Story:** "Alex realized that the hardships faced during that year taught them resilience and adaptability. The obstacles, while difficult, became lessons in patience and problem-solving, paving the way for future success."
 - **Benign Understanding:** Offer compassionate or neutral perspectives on challenging situations. **Example Output in Story:** "As Taylor reflected on the argument, they understood that the other person might have been acting out of stress rather than malice. This realization helped Taylor approach the situation with compassion rather than anger."
 - **Assimilation:** Align situational experiences with broader life meaning systems to foster coherence. **Example Output in Story:** "Though the failure was painful, Emma saw it as a turning point. It reminded her of her commitment to perseverance and the importance of staying true to her passions, even when faced with setbacks."
 - **Downward Comparison:** Emphasize resilience by contrasting the user's challenges with hypothetically more difficult situations. **Example Output in Story:** "Though Chris struggled with balancing work and school, they reflected on how fortunate they were to have access to education and a supportive family. This perspective gave Chris the strength to push through the tough times."

- **Acceptance and Growth:** Highlight personal progress and acceptance of past events while fostering emotional resilience. **Example Output in Story:** *"After months of grappling with loss, Sam found peace in accepting what*

couldn't be changed. Over time, they discovered a deeper appreciation for life and the strength to create new, meaningful connections."

- Use vivid imagery and relatable language to enhance user engagement and connection with the story.